

Quiz 10

STAT 109: Introductory Biostatistics

Quiz 10

Quiz 10 practice problems

These practice problems cover Lecture 25 (one numerical variable across groups: spread, boxplots, group summaries; mean vs median as summaries of center), Lecture 26 (two numerical variables: scatterplots, form / direction / strength), Lecture 27 (two categorical variables: two-way tables, joint / row / column proportions, stacked bar charts), and Lab 10 (missing data, `read_csv`, bivariate plots in R).

Instructions

- Work without notes first, then check your work.
 - For multiple-choice items, choose the best single answer.
 - You may use a scientific calculator on this quiz and your hand draw “methods” table (aka our statistical “toolbox” organized by purpose and type of data).
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Part A: Read the graph (printed figures)

The figures below are for practice only (made-up teaching data). In each part, answer the multiple-choice questions.

A1. Scatterplot (Lecture 26)

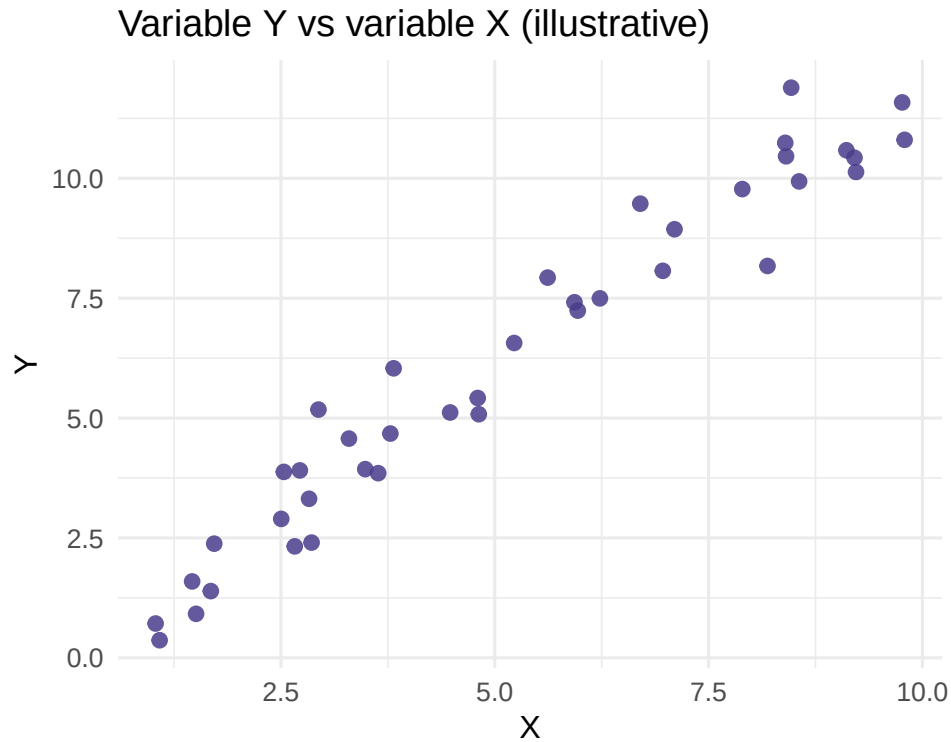


Figure 1: Practice scatterplot for Quiz 10.

(a) Which phrase best describes the **direction** of the association?

- a) Positive (as X increases, Y tends to increase).
- b) Negative (as X increases, Y tends to decrease).
- c) No clear direction.

(b) Which phrase best describes **strength**?

- a) Strong (points hug a clear trend).
- b) Weak (almost no pattern).
- c) Moderate (a visible trend with noticeable scatter).

(c) Which phrase best describes **form**?

- a) Roughly linear.
 - b) Clearly curved (nonlinear).
 - c) No simple form.
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A2. Boxplots comparing two groups (Lecture 25)

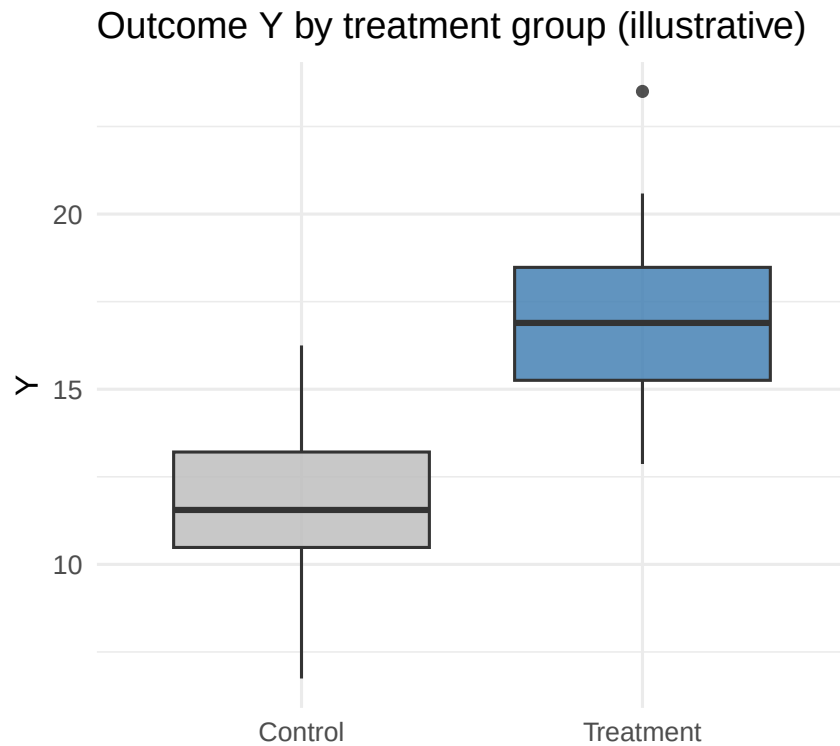


Figure 2: Side-by-side boxplots for two groups.

(a) Which group has the **larger median**?

- a) Control
- b) Treatment
- c) They are essentially the same.

(b) Which group's IQR appears to be larger?

- a) Control
- b) Treatment
- c) Hard to tell from the plot.

A3. Stacked bar chart (Lecture 27)

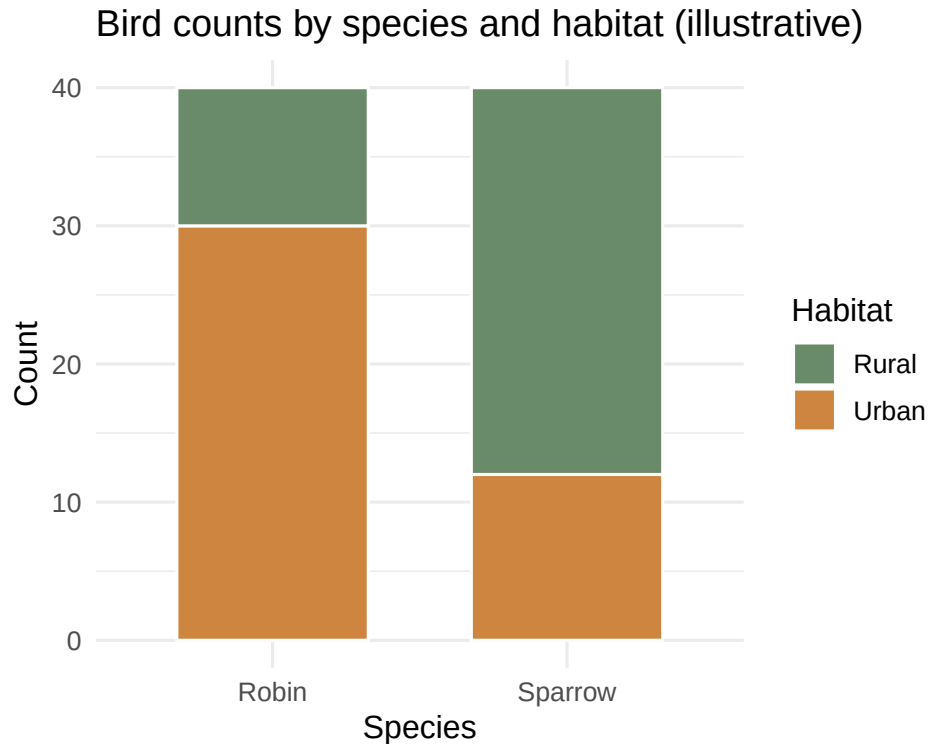


Figure 3: Stacked counts: habitat by species.

Each **bar** is a species; **segments** are Urban vs Rural; **height of a segment** is the count in that cell.

(a) The **total height** of the Robin bar equals which quantity?

- a) The number of Robins (Urban + Rural).
- b) The proportion of all birds that are Robins.
- c) The number of Urban birds of all species.

(b) The **height of the Urban segment** in the Robin bar divided by the **full height of the Robin bar** equals which quantity?

- a) A **joint** proportion (that cell count $\div$ total birds of all species).
- b) A **row** proportion (Urban among Robins).
- c) A **column** proportion (Robins among Urban birds).

A4. Mean vs median — total / per-unit share vs a “typical” case (Lecture 25 / Lecture 24 ideas)

(a) Suppose you are writing a report on that uses data on crop weights harvested from many small plots. Rather than report each individual weight, you want to report one summary number. You think that readers might want to estimate the overall tonnage from a set of plots. Which summary is more appropriate to report?

- a) Median weight of crops from your plots
- b) Mean weight of crops from your plots
- c) The range of weights of crops from your plots

(b) Suppose you are summarizing home sale prices in a city where most houses sell for similar amounts but a few are very expensive. You want a single number that reflects what feels like a typical home price for a buyer, not a number

dominated by those few extremes. Which is usually more appropriate to report?

- a) Mean price.
 - b) Median price.
 - c) The minimum price only.
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Part B: Choose the recommended plot (multiple choice)

For each pair of variables, pick the **best** primary plot from the list (assume you want one clear graphic for a report):

Choices for plotting two variables: (same for each question):

- a) Scatterplot (`geom_point`)
- b) Boxplot per category (`geom_boxplot` with `fill` or `group`)
- c) Histogram with groups overlaid or faceted (`geom_histogram` with `fill` or `group`)
- d) Stacked bar chart (`geom_col` with `fill` or `group`)

B1.

Variable 1: stem diameter (cm)

Variable 2: forest type (pine, oak, mixed)

B2.

Variable 1: soil nitrate (mg/kg)

Variable 2: leaf nitrogen (%)

B3.

Variable 1: island (Newberg, Erie, Coney).

Variable 2: species (blackbeak, redbeak, bluebeak)

B4.

Variable 1: body mass (g)

Variable 2: coast (west, east) and You want to compare the shape of the mass distribution between the two coasts.

Part C: Lab 10 — `read_csv` and `drop_na` (multiple choice)

C1. `read_csv()`

What does `read_csv("https://example.com/data.csv")` (with a valid URL) do in R?

- a) It reads a “colon separated value” file and replaces all the colons with dashes.
 - b) It reads a CSV file from the web into R as a data frame-like object (typically a tibble).
 - c) It deletes all missing values in the dataset and then loads the result as a macaroon.
 - d) It only works on files you already uploaded to Google Drive.
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C2. drop_na()

Suppose we write `penguins |> drop_na(species, bill_length_mm)`. What is the **main effect**?

- a) It drops the columns `species` and `bill_length_mm` from the `penguins` dataset.
 - b) It removes every row that has an NA in either `species` or `bill_length_mm` (or both).
 - c) It replaces any NA in either `species` or `bill_length_mm` with zero.
 - d) It keeps only rows in `penguins` where both variables `species` and `bill_length_mm` are not NA.
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Part D: Two-way table — proportions by hand

D1. Joint proportions

All plants in a small plot are classified by **shade** (Sun, Shade) and **height class** (Short, Tall). The **counts** are:

	Short	Tall	Row total
Sun	6	14	20
Shade	10	10	20
Column total	16	24	40 (grand total N)

Compute the **four joint proportions**. Check that they sum to 1.

D2. Row vs column proportions

Use the same table as in D1 and compute

- (a) Row proportions: what fraction of Sun plants are Short, and what fraction are Tall?
 - (b) Column proportions: what fraction of Short plants are in Sun, and what fraction are in Shade?
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Part E: Boxplot by hand (Lecture 25)

E1. Five-number summary and sketch

The sorted leaf length values (cm) for $n = 9$ plants are:

2, 4, 5, 6, 7, 8, 9, 11, 15

Use the same splitting rule as in Lecture 25 for odd n : the median is the middle value; the lower half and upper half for quartiles exclude the median (four values in each half); for four numbers, Q_1 and Q_3 are the averages of the two middle values in each half.

- (a) Report the five-number summary: min, Q_1 , median, Q_3 , max.

(b) On paper, sketch a boxplot of these data (remember this is a box from Q_1 to Q_3 , median line inside, whiskers to min and max (no outlier rules for this exercise). Label the numeric positions of the box edges and median on the axis.